Product Requirements Document

Mini AI Voice KAM: a voice-first Virtual Key Account Manager for Snapdeal sellers

Version	Status	Owner	Deadline
v1.0	Draft for review	Candidate (AI First Intern)	Within 24 h of receiving the brief

1. Summary

A seller calls a Virtual KAM because sales have dropped. We will build a **browser-based voice agent** that greets the seller, listens, asks focused diagnostic questions, reads the seller's metrics, identifies the root cause, recommends **2–3 practical actions** and ends with a spoken and on-screen **summary of issues, actions and next steps**.

The guiding idea comes from the hiring email: the agent should not just answer questions. It should **take ownership** of the seller's problem. It comes to the call already having read the data, forms a hypothesis, checks it with the seller and drives the next action.

2. Problem statement

Sellers usually see that sales fell but not *why*. Human KAMs cannot cover the long tail of small sellers, so most of them get no timely, data-backed guidance. A Virtual KAM that turns raw metrics into a clear cause and a short action plan, over a natural voice call, closes this gap at scale.

> **Scenario given in the brief**
> “My sales have dropped significantly this month. I don’t understand why. Can you help me?”

3. Target user

Attribute	Assumed profile
Who	Small or mid-size Snapdeal seller who runs their own inventory and listings
Context	Busy, may be calling from a phone or browser, wants a fast answer and not a report
Data literacy	Knows their products well; less comfortable reading dashboards and metrics
Needs	A plain-language reason for the drop, actions they can take this week, and someone who follows up

4. Goals and non-goals

Goals

- Hold a natural, two-way voice conversation that the seller can interrupt (rubric: 30 marks).
- Understand the seller's concern and ask relevant follow-up questions (20 marks).
- Diagnose the decline from the sample data and give 2–3 specific, usable recommendations (20 marks).
- Use a clean, explainable AI workflow where numbers come from code and not from the LLM (20 marks).
- Add a few touches that show product thinking: a live insights panel and a call summary (10 marks).

Non-goals

- Real Snapdeal data, systems, authentication or integrations (the brief rules these out).
- Telephony / phone numbers, mobile apps, multi-language support (English only for v1).
- Actually executing changes on the seller's account. Actions are recorded as an action plan.

- A polished UI. The interface stays minimal and functional.

5. Data analysis

The brief supplies one month-over-month snapshot. We analyse it first, because the quality of the bot's advice depends entirely on reading this table correctly.

Metric	Last month	This month	Change	What it tells us
Sales	₹2,00,000	₹1,35,000	**−32.5%**	Revenue fell by ₹65,000
Orders	400	270	**−32.5%**	Same % as sales, so the whole drop is volume
Average selling price	₹500	₹500	0%	Pricing did **not** change; price is not the cause
Cancellation rate	8%	14%	**+6 pts (1.75×)**	32 → ~38 cancelled orders, even though orders fell
Out-of-stock products	5	18	**+13 (3.6×)**	Big inventory gap; the strongest signal in the data

5.1 Root-cause hypothesis: one inventory problem, three symptoms

All three bad metrics can be explained by a single cause: **stock-outs**.

- **Fewer live listings → fewer orders.** Out-of-stock products cannot be bought and are typically pushed down or hidden in search. With 13 more products unavailable, the seller simply had less to sell.
- **Stock running out after orders are placed → more cancellations.** When inventory is not kept in sync, customers order items the seller no longer has, and the seller cancels.
- **Cancellations and stock-outs → lower visibility.** Marketplaces generally rank reliable sellers higher. A rising cancellation rate can reduce ranking even for in-stock products, so the drop can feed itself.

> **Key insight the bot should explain to the seller**
> Under assumption A3 (a catalogue of ~50 products), live products fell from 45 to 32 (−29%), close to the −32.5% drop in orders. Orders *per live product* stayed roughly flat (8.9 → 8.4). In other words, demand for each listed product held up; the seller lost sales mainly because fewer products were available to buy.

5.2 Size of the opportunity (estimates, under stated assumptions)

Lever	Estimate	Basis
Restock the 13 extra out-of-stock products	≈ 110–115 orders ≈ **₹55k–58k / month**	13 products × ~8.4–8.9 orders per live product (section 5.1)
Bring cancellations back from 14% to 8%	≈ 16 orders ≈ **₹8k / month**	6% of 270 orders × ₹500
Delivered value, for context	₹1,84,000 → ₹1,16,100 (**−37%**)	Orders × (1 − cancellation rate) × ₹500: the realised drop is worse than headline sales

6. Assumptions (stated openly, as the brief requires)

ID	Assumption	Why it is reasonable / how it is used
A1	“Sales” is gross order value (orders × ASP), before cancellations.	400 × ₹500 = ₹2,00,000 and 270 × ₹500 = ₹1,35,000 match exactly.
A2	Order counts include orders that were later cancelled.	Follows from A1; used to estimate delivered value.
A3	The seller lists ~50 active products in one category (Home & Kitchen).	Not given in the brief; needed to reason about the share of the catalogue that is unavailable. The bot calls this an estimate.

ID	Assumption	Why it is reasonable / how it is used
A4	Some of the 18 out-of-stock products were recent best-sellers.	Typical pattern: fast movers run out first. A small mock product list illustrates this.
A5	Most of the extra cancellations are seller-initiated, caused by stock running out.	Treated as a hypothesis. The bot **asks the seller to confirm**, it does not assert it.
A6	Snapdeal-style marketplace mechanics: out-of-stock listings lose visibility and high cancellation rates hurt seller ranking.	General marketplace behaviour; described as typical, not as Snapdeal policy.

Extended mock data (illustrative and clearly labelled in the app): seller name, category and a short list of top out-of-stock products with last month's units sold. This lets the bot make concrete suggestions such as “restock these three first”.

7. Functional requirements

FR1–FR6 map one-to-one to section 3 of the brief. FR7–FR8 are our additions.

ID	Requirement	Acceptance criteria
FR1	**Greet and understand intent.** Warm greeting as the seller's Virtual KAM; asks how it can help.	Greeting within ~2 s of starting the call; correctly identifies “sales dropped” intent from free speech.
FR2	**Natural two-way voice.** Speech in, speech out, with turn-taking and interruption (barge-in).	Seller can interrupt the bot mid-sentence; replies are short (1–3 sentences); no typing needed.
FR3	**Relevant follow-up questions.** Hypothesis-driven, one question at a time.	Asks at least 2 questions that test the hypothesis (e.g. which products ran out and why; who is cancelling orders).
FR4	**Use seller data to find issues** (pricing, inventory, cancellations, visibility).	Bot fetches metrics via a tool; every number it speaks matches the data; explicitly rules out pricing.
FR5	**Recommend 2–3 practical actions.**	Each action is specific, has a reason tied to the data and a timeframe.
FR6	**Summarise at the end:** key issues, recommendations, next steps.	Spoken summary of ≤ 30 s plus the same summary shown on screen.
FR7	**Live insights panel** (addition).	Page shows the metrics table and the findings once the bot has analysed the data.
FR8	**Ownership and follow-up** (addition).	Bot proposes a concrete follow-up (e.g. check-in after 7 days with the metrics to watch).

8. Conversation design

Stage	Bot behaviour	Example line (illustrative)
1. Greet	Introduce itself as the seller's Virtual KAM; open question.	“Hi, I’m your Snapdeal account manager. How can I help today?”
2. Acknowledge	Show empathy, confirm the problem, say it has already looked at the account.	“I can see that. Your orders fell from 400 to 270 this month. Let’s find out why.”
3. Diagnose	Share the main signal, then ask one question at a time to confirm it.	“Your price is unchanged, but 18 products are out of stock compared with 5 last month. Did you have a supply problem?”
4. Probe	Ask about cancellations and their cause; adapt to the seller's answers.	“Cancellations went up to 14%. Are those mostly orders you couldn’t fulfil?”
5. Explain	Connect the dots in plain language (one cause, three symptoms).	“So stock-outs are hurting you twice: fewer products to sell, and cancellations that lower your ranking.”
6. Recommend	2–3 prioritised actions with the expected impact.	See section 9.

Stage	Bot behaviour	Example line (illustrative)
7. Summarise and commit	Recap issue, actions, next step; propose a follow-up; close politely.	“To recap… I’ll check in next week to see if orders are recovering.”

Conversation rules

- Speak like a person on a phone call: short sentences, no lists or symbols read aloud, numbers said naturally.
- One question per turn; always wait for the seller's answer.
- Never invent numbers. Only use values returned by the data tool, and say so when something is an estimate.
- If the seller's answer contradicts the hypothesis (e.g. stock is fine), revise the diagnosis instead of insisting.
- Politely steer off-topic requests back to the sales problem; say “I don’t have that data” when true.

9. Recommendation playbook

The bot picks and orders recommendations based on the diagnosis and the seller's answers. For the sample data the expected set is:

#	Action	Why (tied to data)	When / impact
R1	**Restock the out-of-stock products, best-sellers first**, and set a reorder alert (e.g. at ~2 weeks of stock cover).	18 products out of stock vs 5; orders per live product held steady, so restocking should bring orders back.	Within 48–72 h; largest lever, ≈ ₹55k–58k / month.
R2	**Stop cancellations at the source:** update stock counts daily, deactivate or pause listings at zero stock, keep a small buffer.	Cancellations up from 8% to 14%, likely from selling stock that was not there.	This week; ≈ ₹8k / month and protects ranking.
R3	**Rebuild visibility after restocking:** short promotion or sponsored listing on restocked best-sellers; join an upcoming sale event if eligible.	Stock-outs and cancellations likely pushed listings down; ranking recovers faster with a sales push.	After restock; speeds up recovery.

Explicitly **not** recommended: cutting prices. The average selling price is unchanged, so a discount would reduce margin without fixing the cause. Saying this shows the bot reasons rather than giving generic advice.

10. Non-functional requirements

Area	Requirement
Latency	Bot starts speaking within ~1.5 s of the seller finishing a turn (perceived).
Naturalness	Natural English voice; barge-in supported; filler-free, concise answers.
Accuracy	Zero invented figures: all metrics come from the data tool; estimates labelled as such.
Security	ElevenLabs API key stays on the server (.env); the browser gets a short-lived session token only.
Reliability	If the data tool fails, the bot says so and continues with general guidance instead of crashing.
Simplicity	One Python service plus one HTML page; runs locally with one command; deployable for a demo link.

11. Solution overview

Full detail goes in the technical note. At a high level:

Component	Choice	Role
Voice agent	ElevenLabs Agents (Conversational AI)	Speech-to-text, turn-taking, interruption handling, LLM orchestration, text-to-speech
LLM	Selected inside the ElevenLabs agent (finalised during build)	Conversation, reasoning, choosing when to call tools

Component	Choice	Role
Backend	Python + FastAPI	Serves the page, issues secure session tokens, hosts the seller-data and diagnosis tool
Diagnosis engine	Plain Python rules	Computes changes, ranks likely causes and estimates impact. Deterministic and testable
Frontend	Single HTML/JS page	Start/stop call, live transcript, metrics and findings panel, end-of-call summary

Key design decision: numbers come from code, conversation comes from the LLM
The LLM is good at conversation and judgement but unreliable at arithmetic. So the Python diagnosis engine computes every number and ranks causes, and the LLM only explains and discusses the results. This makes the bot's advice accurate, testable and easy to explain.

12. How the solution maps to the evaluation rubric

Criterion (marks)	How we address it
Voice interaction & natural conversation (30)	Managed real-time voice with barge-in; prompt tuned for short, spoken-style turns; tested for latency.
Understanding seller queries (20)	Intent recognition from free speech; hypothesis-driven follow-ups; adapts when the seller disagrees.
Quality of business recommendations (20)	Root cause linking all three symptoms; 2–3 prioritised actions with reasons, timeframes and ₹ impact; explains why a price cut is not advised.
AI implementation & workflow (20)	Clear separation: tool-based data access, deterministic diagnosis, LLM for dialogue; secure key handling.
Creativity & execution (10)	Live insights panel, on-screen call summary, proactive follow-up commitment.

13. Test plan (acceptance scenarios)

#	Scenario	Expected behaviour
T1	Happy path: seller states the brief's line and answers questions cooperatively.	Identifies stock-outs as the root cause, gives R1–R3, ends with a summary.
T2	Seller interrupts the bot mid-sentence.	Bot stops, listens, responds to the interruption.
T3	Seller says “my stock is fine”.	Bot does not insist; probes cancellations and visibility instead.
T4	Seller asks “should I reduce my prices?”	Explains that price is unchanged and a cut would not fix the cause.
T5	Off-topic question (e.g. payment settlement).	Acknowledges, says it has no data on that, steers back.
T6	Number check across the whole call.	Every figure spoken matches the data or is labelled an estimate.

14. Risks and mitigations

Risk	Mitigation
ElevenLabs free-tier minutes run out during testing	Test logic in text first; keep voice tests short; record the demo early.
Speech recognition mishears numbers or “₹”	Bot reads numbers from the tool, not from the transcript; confirms key facts.
LLM states made-up figures	Numbers only via tool; prompt forbids invented data; test T6.
Robotic or long-winded replies	Prompt rules for short spoken turns; iterate on real test calls.

Risk	Mitigation
Microphone blocked in browser	Serve over HTTPS when deployed; clear on-screen permission prompt.
Time pressure (24 h)	Build the minimum end-to-end first; extras (panel, summary card) only after it works.

15. Deliverables

Deliverable	Required by brief?	Format
Working demo link or runnable prototype	Yes	Deployed URL + GitHub repo with README
1–2 minute screen recording of a real voice call	Yes	MP4
Short note (≤ 1 page): models, voice platform, workflow	Yes	PDF
This PRD	No (supporting)	PDF
Interview prep book: every decision and the reasoning behind it	No (for the candidate)	PDF
Demo script	No (supporting)	PDF

16. Open questions

- Which LLM inside the ElevenLabs agent gives the best latency/quality balance? Decide by quick A/B test during build.
- Hosting choice for the demo link (e.g. Render vs Railway). Decide at deployment.
- Which voice best fits an Indian seller audience while staying in English? Pick after listening to 2–3 options.